Politecnico di Torino

Advanced Machine Learning

A.A. 2023/2024

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Summary

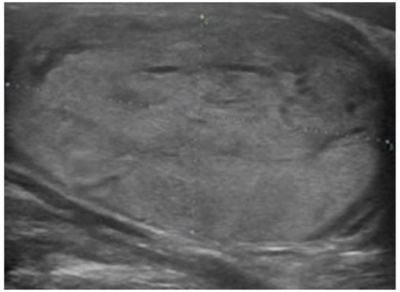
- Learning can only be Probably and Approximately Correct (PAC)
    - true risk vs empirical risk
    - loss
    - empirical risk minimization can lead to learning algorithms with reasonable generalization guarantees (within some conditions)

- Underfitting / Overfitting
    - cross-validation

- Linear Regression
    - ERM with square loss over linear predictors is efficient: just invert a matrix
    - LS learning with basis functions (e.g. polynomials) is still linear
    - without and with regularization

Use a Regression Model for Classification

What if we want to predict a category instead of a value?

$h_\Theta($  $) = \{0, 1\}$

Possible solution: Do post-processing (e.g. thresholding) to convert linear regression to a binary output.

$\Rightarrow$ The solution is not necessarily an optimum anymore.

Instead: Modify the loss to minimize over categorical values directly.

Slide Credit: E. Rodolà

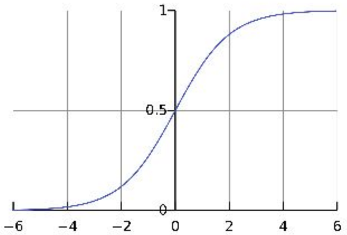
Logistic Regression

New loss:

$$\ell_\Theta(\{x_i, y_i\}) = \sum_{i=1}^{n} (y_i - \sigma(\underbrace{ax_i + b}_{\text{linear}}))^2$$

Here, σ is the nonlinear logistic sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



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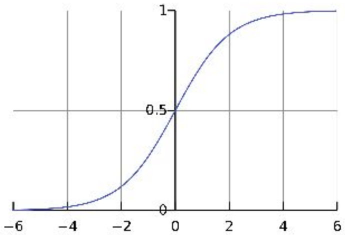

σ has a saturation effect as it maps $\mathbb{R} \mapsto (0, 1)$.

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$$\ell_\Theta(\{x_i, y_i\}) = \sum_{i=1}^{n} (y_i - \sigma(\underbrace{ax_i + b}_{\text{linear}}))^2 \qquad \text{non-convex}$$

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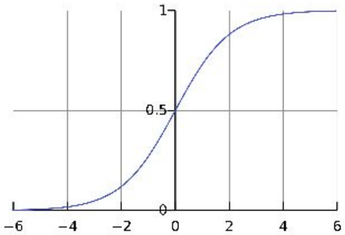

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$$c(x_i, y_i) = \begin{cases} -\ln(\sigma(ax_i + b)) & y_i = 1 \\ -\ln(1 - \sigma(ax_i + b)) & y_i = 0 \end{cases} \qquad \text{convex}$$

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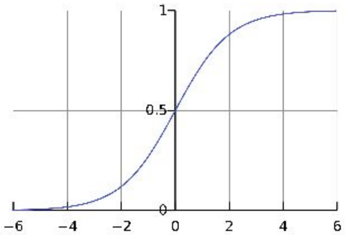

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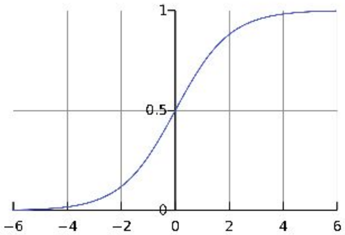

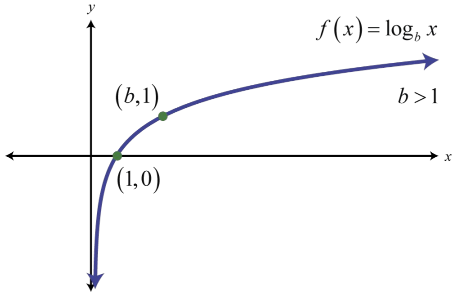

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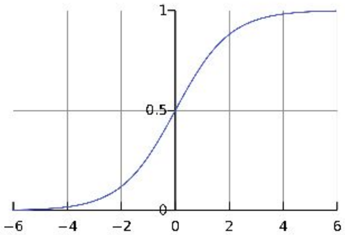

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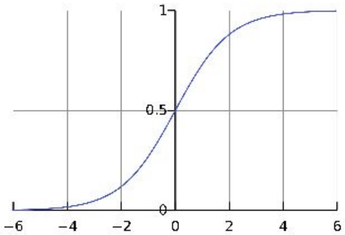

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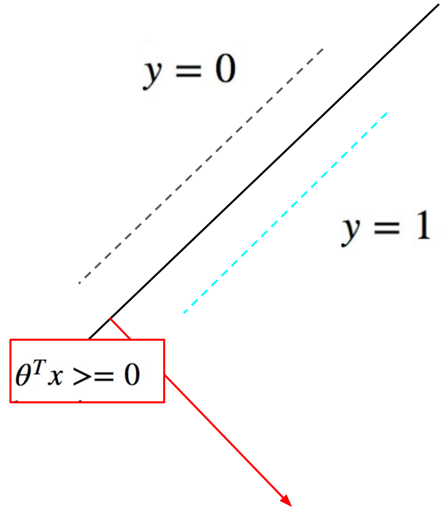
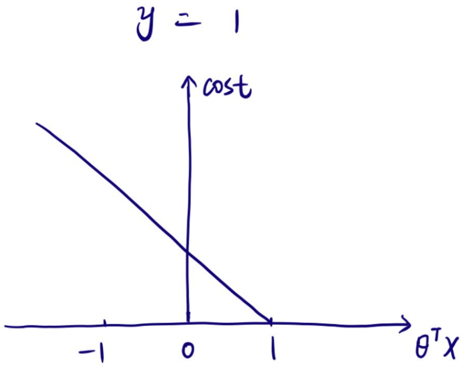
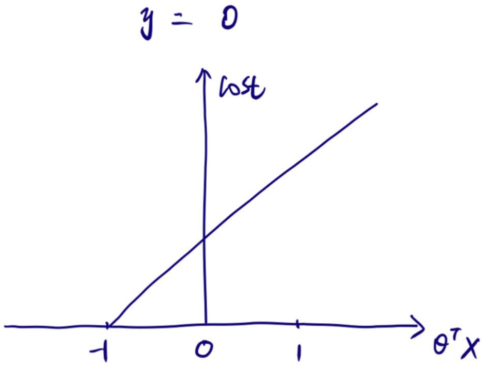
What was the loss in SVM?



$$h_\theta(x) = \begin{cases} 1 & \text{if } \theta^T x >= 0 \\ 0 & \text{otherwise} \end{cases}$$

$$Cost(h_\theta(x), y) = \begin{cases} max(0, 1 - \theta^T x) & \text{if } y = 1 \\ max(0, 1 + \theta^T x) & \text{if } y = 0 \end{cases}$$





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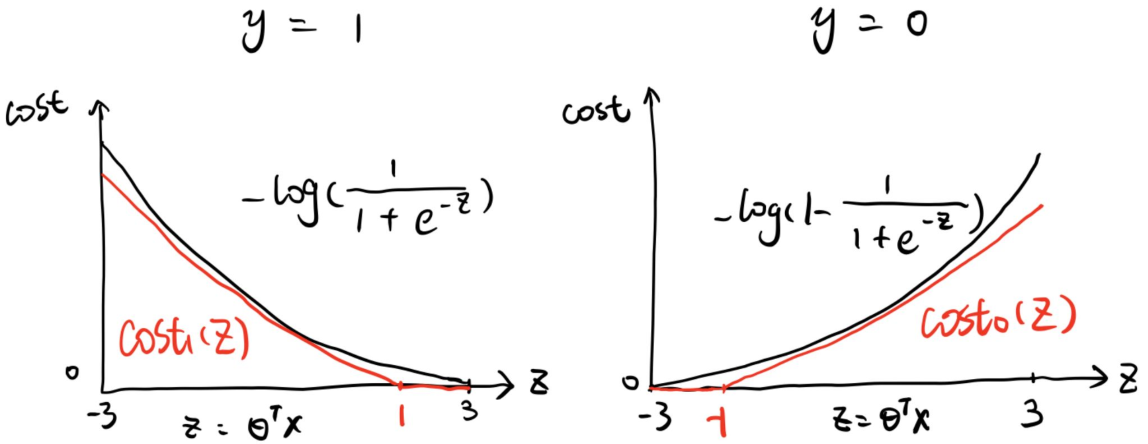

https://towardsdatascience.com/optimization-loss-function-under-the-hood-part-iii-5dff33fa015d

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$$J(\theta) = \sum_{i=1}^{m} y^{(i)} Cost_1(\theta^T(x^{(i)})) + (1 - y^{(i)}) Cost_0(\theta^T(x^{(i)}))$$

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$$J(\theta) = C[\sum_{i=1}^{m} y^{(i)} Cost_1(\theta^T(x^{(i)}) + (1 - y^{(i)}) Cost_0(\theta^T(x^{(i)})] + \frac{1}{2}\sum_{j=1}^{n} \theta_j^2$$

$$m = number\ of\ samples, \quad n = number\ of\ features$$

AIML

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Logistic Regression

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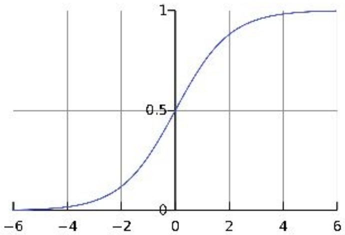

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Logistic Regression: Finding a Solution

Since the loss is convex, the first-order conditions apply:

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Consider the gradient of each term in the summation:

$$\nabla_\Theta \left(y_i \ln(\sigma(ax_i + b)) + (1 - y_i) \ln(1 - \sigma(ax_i + b)) \right)$$

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$$y_i \nabla_\Theta \underbrace{\ln(\sigma(ax_i + b))}_{f(g(h(\Theta)))} + (1 - y_i) \nabla_\Theta \ln(1 - \sigma(ax_i + b))$$

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Apply the chain rule to each partial derivative:

$$\frac{\partial}{\partial a} f(g(h(a, b))) = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial h}{\partial a}$$

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$$\boxed{\sigma(x) = \frac{1}{1 + e^{-x}}}$$

$$= \frac{\partial f}{\partial g} \cdot \frac{\partial \sigma(ax_i + b)}{\partial(ax_i + b)} \cdot x_i$$

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$$\begin{aligned}
\frac{\partial}{\partial a} f(g(h(a,b))) &= \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial h}{\partial a} \quad = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial}{\partial a} ax_i + b \\
&= \frac{\partial f}{\partial g} \cdot \frac{\partial \sigma(ax_i + b)}{\partial(ax_i + b)} \cdot x_i \quad = \frac{\partial f}{\partial g} \cdot \frac{\partial}{\partial(ax_i + b)} \frac{1}{1 + e^{-(ax_i + b)}} \cdot x_i \\
&= \frac{\partial f}{\partial g} \cdot \frac{e^{-(ax_i + b)}}{(1 + e^{-(ax_i + b)})^2} \cdot x_i
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Logistic Regression: Finding a Solution

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$$= \frac{\partial f}{\partial g} \cdot \frac{1}{1 + e^{-(ax_i+b)}} \frac{(1 + e^{-(ax_i+b)}) - 1}{1 + e^{-(ax_i+b)}} \cdot x_i \quad = \frac{\partial f}{\partial g} \cdot \frac{1}{1 + e^{-(ax_i+b)}} (1 - \frac{1}{1 + e^{-(ax_i+b)}}) \cdot x_i$$

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&= \frac{\partial f}{\partial g} \cdot \sigma(ax_i + b)(1 - \sigma(ax_i + b)) \cdot x_i
\end{aligned}$$

Logistic Regression: Finding a Solution

$$y_i \nabla_\Theta \underbrace{\ln(\sigma(ax_i + b))}_{f(g(h(\Theta)))} + (1 - y_i)\nabla_\Theta \ln(1 - \sigma(ax_i + b))$$

Apply the chain rule to each partial derivative:

$$\frac{\partial}{\partial a} f(g(h(a, b))) = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial h}{\partial a} \quad = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial}{\partial a} ax_i + b$$

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Slide Credit: E. Rodolà

Logistic Regression: Finding a Solution

Since the loss is convex, the first-order conditions apply:

$$\nabla_\Theta \sum_{i=1}^{n} y_i \ln(\sigma(ax_i + b)) + (1 - y_i) \ln(1 - \sigma(ax_i + b)) = 0$$

where $\Theta = \{a, b\}$.

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Apply the chain rule to each partial derivative:

$$\frac{\partial}{\partial a} \ln(\sigma(ax_i + b)) = (1 - \sigma(ax_i + b)) x_i$$

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$$y_i \nabla_\Theta \underbrace{\ln(\sigma(ax_i + b))}_{f(g(h(\Theta)))} + (1 - y_i)\nabla_\Theta \ln(1 - \sigma(ax_i + b))$$

Apply the chain rule to each partial derivative:

$$\frac{\partial}{\partial a} \ln(\sigma(ax_i + b)) = (1 - \sigma(ax_i + b))x_i$$

And similarly for the second term and for all parameters.

Slide Credit: E. Rodolà

Logistic Regression: Finding a Solution

By looking at the partial derivative:

$$\frac{\partial}{\partial a} \ln(\sigma(ax_i + b)) = (1 - \sigma(ax_i + b))x_i$$

we see that the parameters enter the gradient in a nonlinear way.

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we see that the parameters enter the gradient in a nonlinear way.

Thus:

- $\nabla \ell_\Theta = 0$ is not a linear system to solve for the parameters.
- $\nabla \ell_\Theta = 0$ is a transcendental equation $\Rightarrow$ no analytical solution.

model	loss	solution
linear regression	convex	least squares
logistic regression	convex	nonlinear optimization

Gradient Descent: Intuition

Gradient descent is a first-order iterative minimization algorithm.

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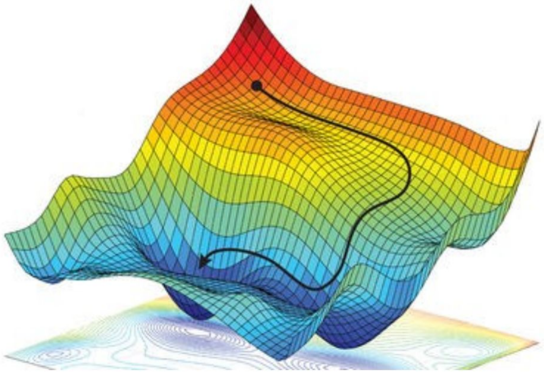
Gradient descent is a first-order iterative minimization algorithm.

Example of a loss function $\ell_{\boldsymbol{\Theta}} : \mathbb{R}^2 \to \mathbb{R}$:



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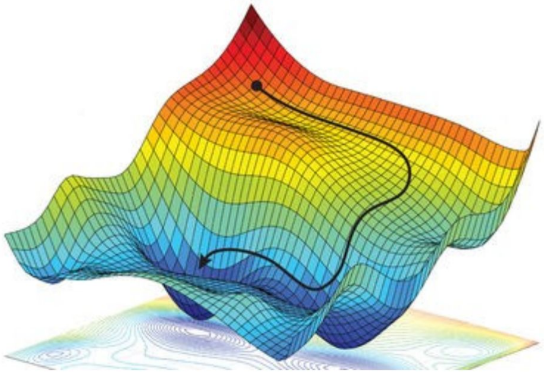


Overall idea: Move where the function decreases the most.

1. Start from some point $\boldsymbol{\Theta}^{(0)} \in \mathbb{R}^2$.
2. Iteratively compute:

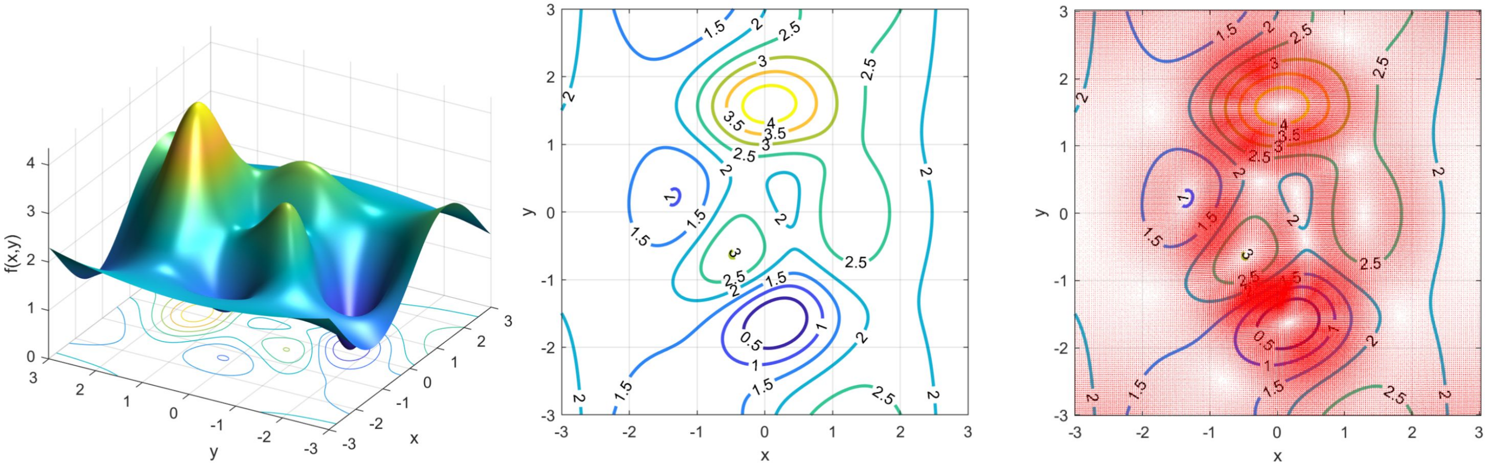
$$\boldsymbol{\Theta}^{(t+1)} = \boldsymbol{\Theta}^{(t)} - \alpha \nabla \ell_{\boldsymbol{\Theta}^{(t)}}$$

3. Stop when a minimum is reached.

Slide Credit: E. Rodolà

Gradient Descent

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$



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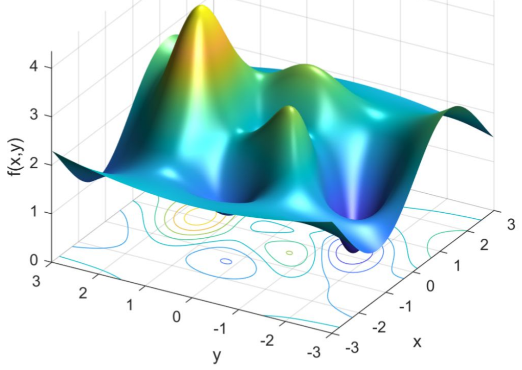

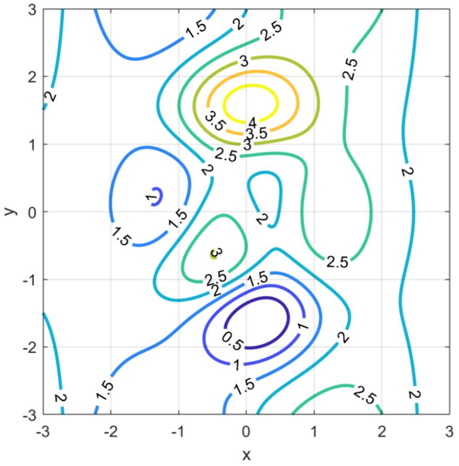

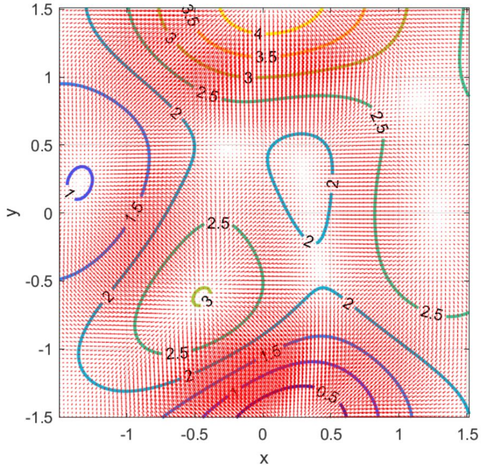

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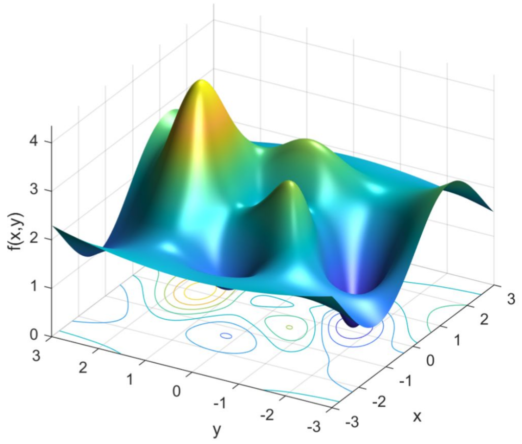

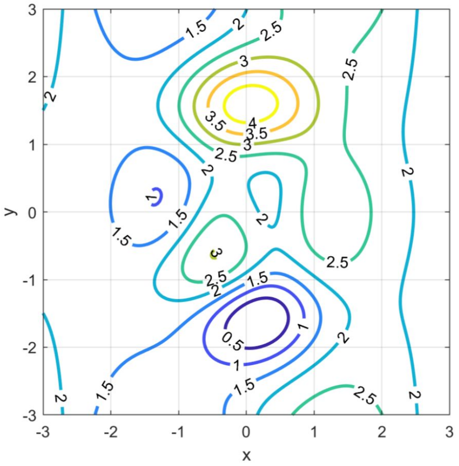

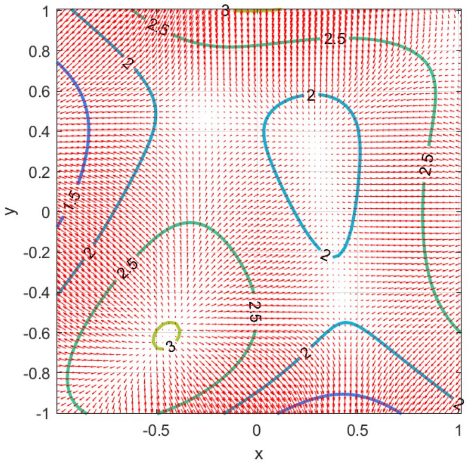

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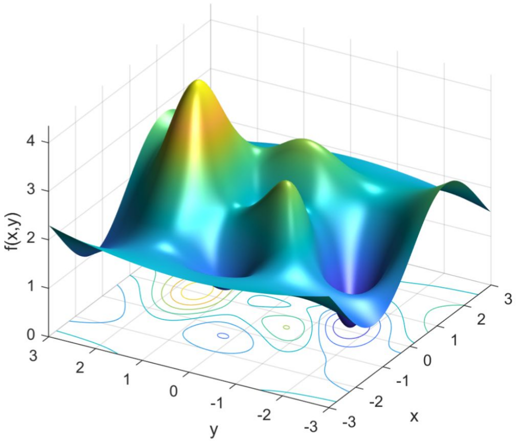

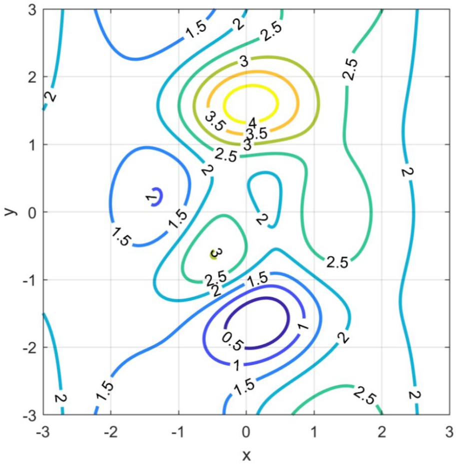

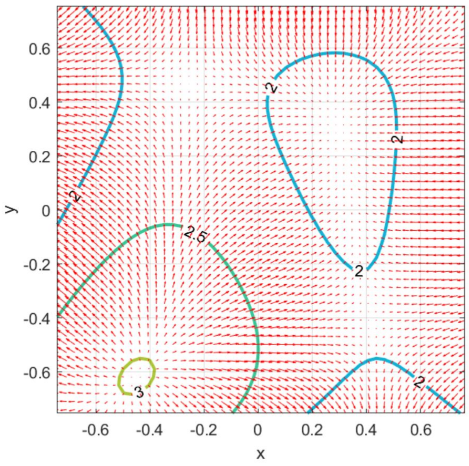

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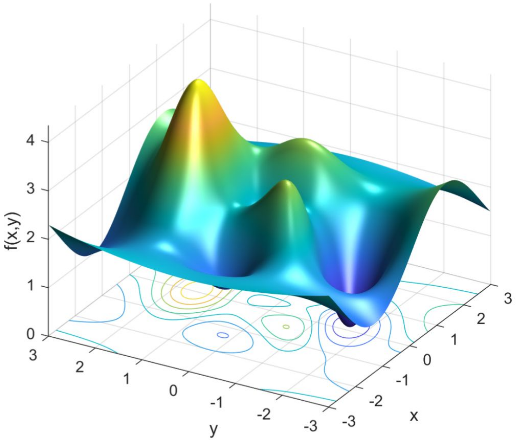

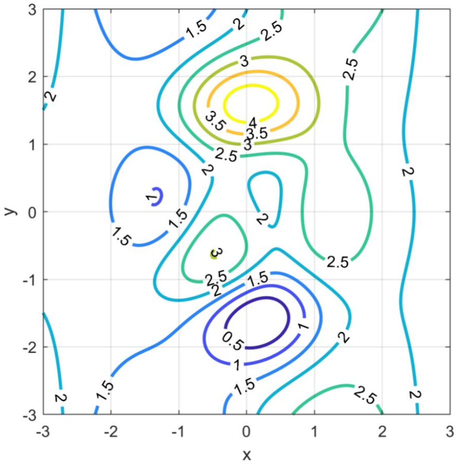

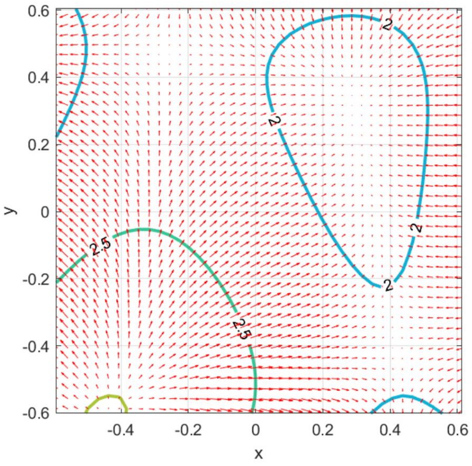

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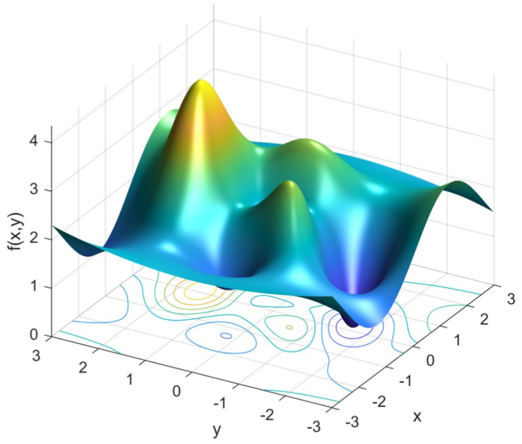
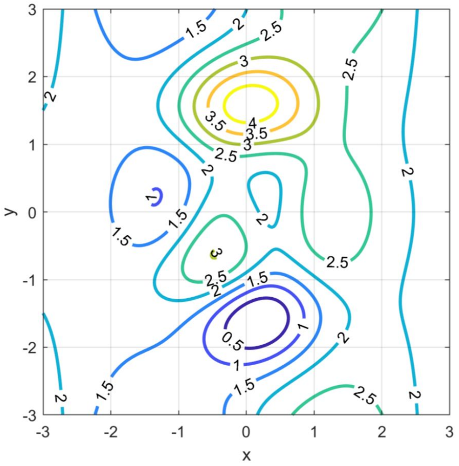
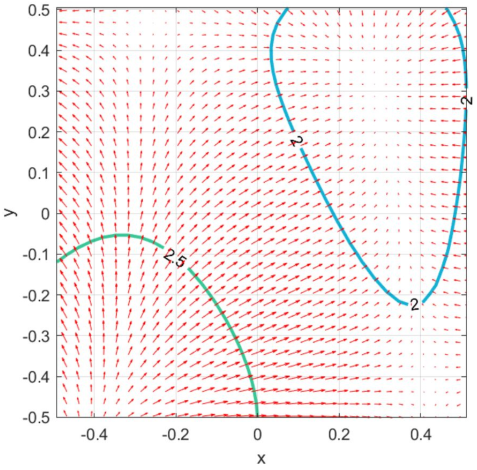


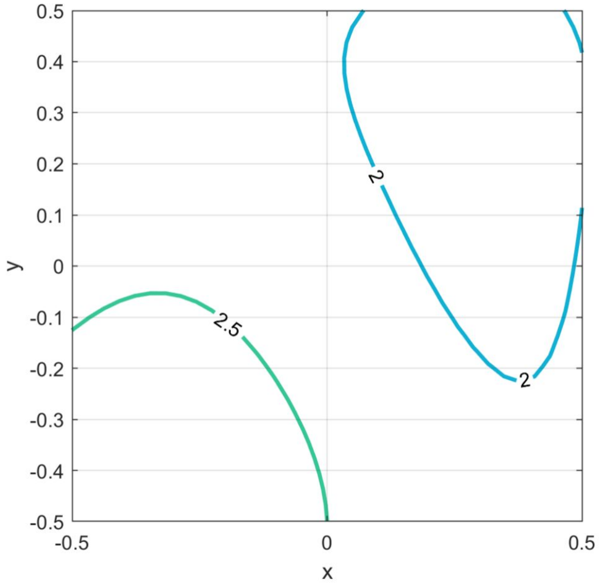




Gradient Descent

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$

The gradient is orthogonal to level curves / level surfaces.



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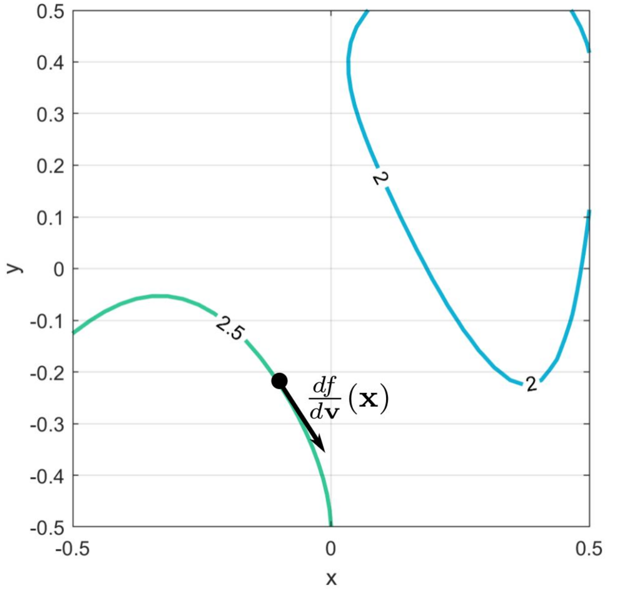

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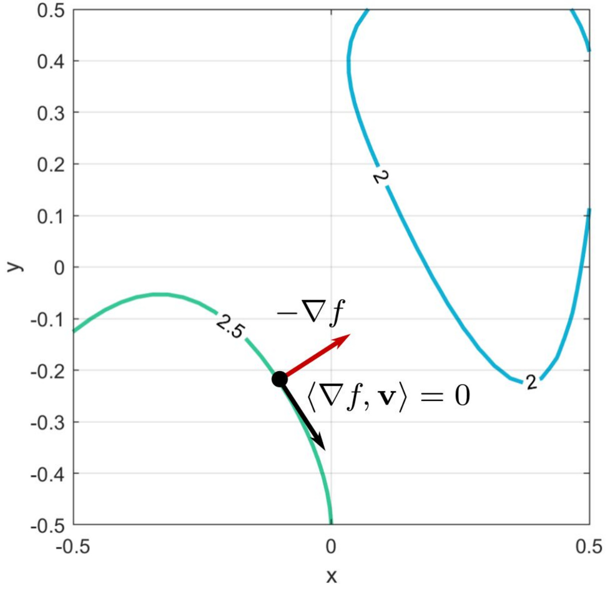


The directional derivative is zero along isocurves.

Gradient Descent: Differentiability

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$

Gradient descent requires f to be differentiable at all points.

Recall:

$$f \text{ has partial (or even directional) derivatives} \not\Rightarrow f \text{ is differentiable}$$

$$f \text{ has continuous gradient} \Rightarrow f \text{ is differentiable}$$

Gradient Descent: Differentiability

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$

Gradient descent requires f to be differentiable at all points.

Example in $\mathbb{R}^2$: $f(x,y) = \begin{cases} 0 & \text{at } (0,0) \\ \frac{x^2 y}{x^2+y^2} & \text{elsewhere} \end{cases}$

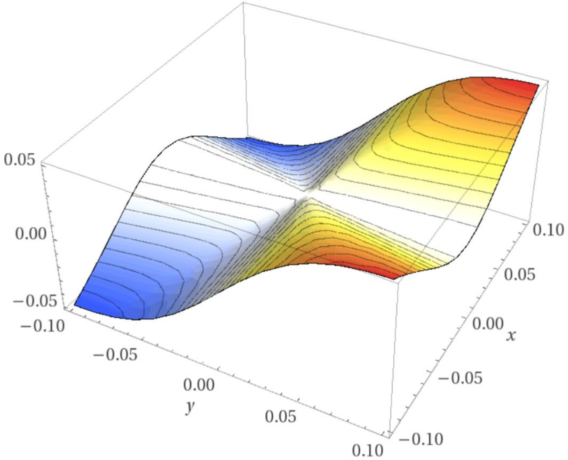
$\frac{\partial}{\partial y} f(x,0) = 1$ for $x \neq 0$

$\frac{\partial}{\partial y} f(0,y) = 0$ for $y \neq 0$

$\Rightarrow \frac{\partial}{\partial y} f(x,y)$ discontinuous at $(0,0)$

the partial derivatives of f are defined everywhere, but are discontinuous at the origin

$\Rightarrow f$ is not differentiable



Gradient Descent: Stationary Points

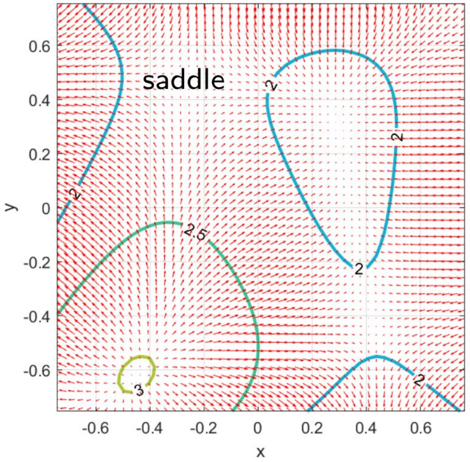
A stationary point is such that:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \cancel{\nabla f(\mathbf{x}^{(t)})} \rightarrow 0$$

Gradient descent "gets stuck" at stationary points.

However:

- Stationary point $\not\Rightarrow$ local minimum



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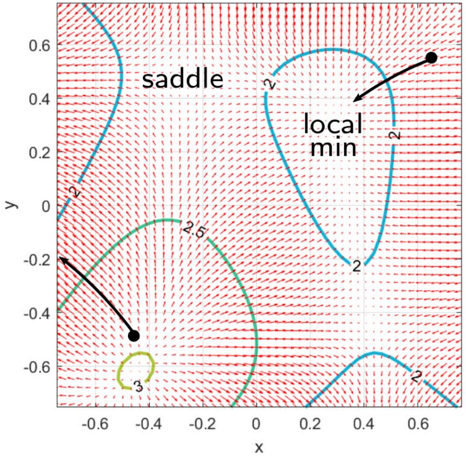
However:

- Stationary point $\not\Rightarrow$ local minimum
- Which stationary point depends on the initialization.



Gradient Descent: Learning Rate

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$

The parameter $\alpha > 0$ is also called learning rate in ML.

Remark: The length of a step is not simply α, but $\alpha \|\nabla f\|$.

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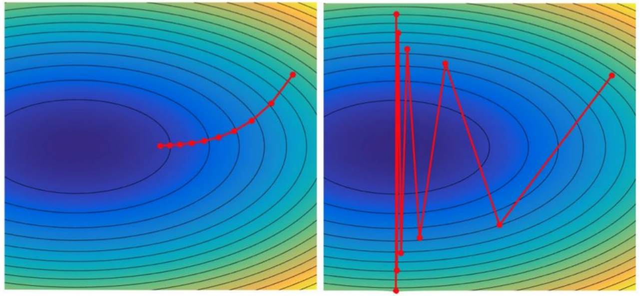
- Too small: slow convergence speed
- Too big: risk of overshooting
- Optimal values can be found via line search algorithms



small α large α

Slide Credit: E. Rodolà

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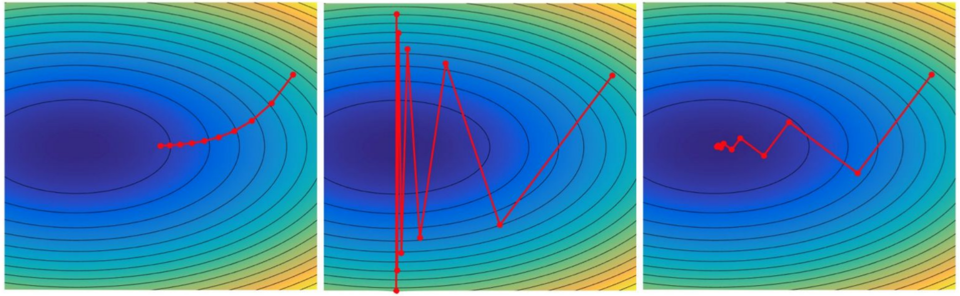

small α large α optimal α

$\arg\min_\alpha f(\mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)}))$

find the alpha that allows you to get to the lowest point of the function where your gradient direction is tangent to an isocurve

Slide Credit: E. Rodolà

Decay and Momentum

The learning rate can be adaptive or follow a schedule.

- Decrease α according to a decay parameter ρ:

  Examples:

$$\alpha^{(t+1)} = (1-\frac{t}{\rho})\alpha^{(0)}+\frac{t}{\rho}\alpha^{(\rho)}\,, \quad \alpha^{(t+1)} = \frac{\alpha^{(t)}}{1+\rho t}\,, \quad \alpha^{(t+1)} = \alpha^{(0)}e^{-\rho t}$$

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- Accumulate past gradients and keep moving in their direction:

$$\mathbf{v}^{(t+1)} = \lambda \mathbf{v}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)}) \quad \text{momentum}$$
$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} + \mathbf{v}^{(t+1)}$$

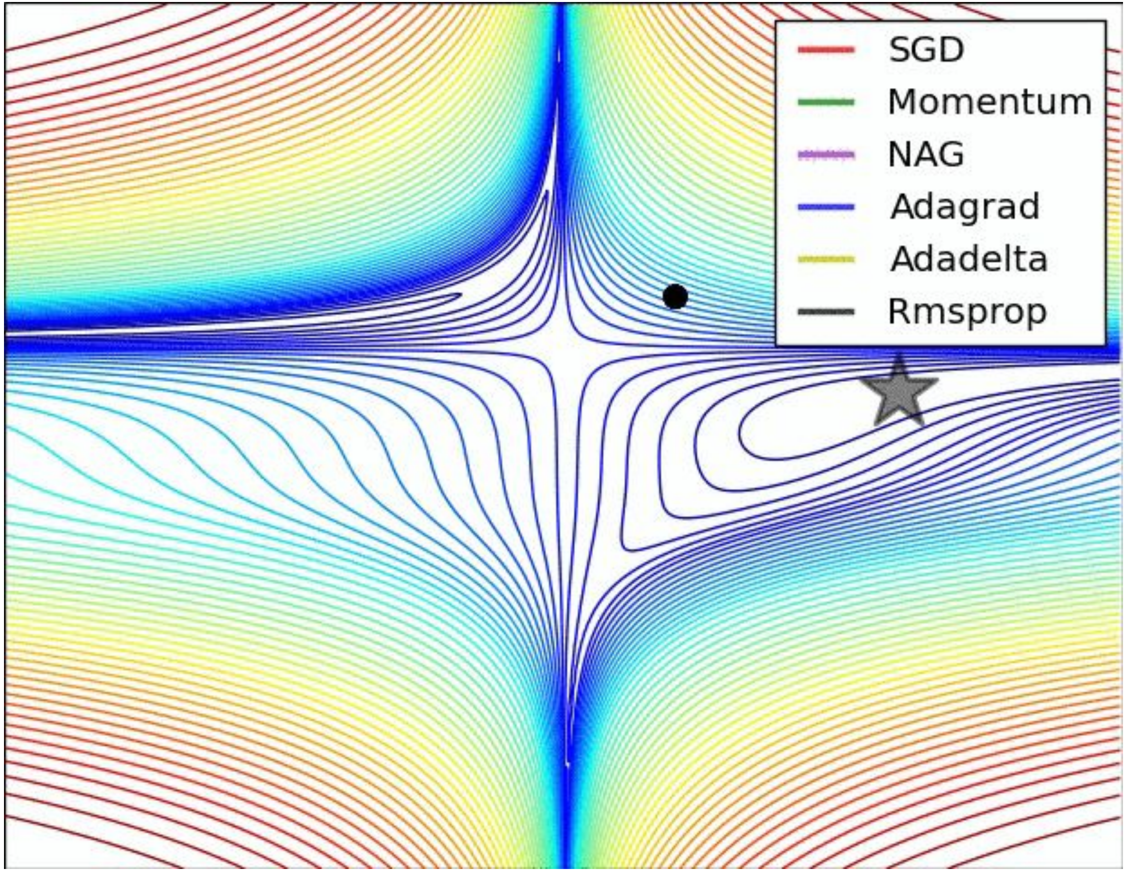
Step length $\propto$ how aligned is the sequence of gradients.

$$\frac{1}{1-\lambda}\alpha \|\nabla f\|$$

Acceleration effect for big λ + escape from local minima.

AI2ML

Momentum



https://cs231n.github.io/neural-networks-3/

First Order Acceleration Methods

Let us try to unroll gradient descent:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \alpha \nabla f(\mathbf{x}^{(t)})$$

$$\begin{aligned}
\mathbf{x}^{(1)} &= \mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)}) \\
\mathbf{x}^{(2)} &= \mathbf{x}^{(1)} - \alpha \nabla f(\mathbf{x}^{(1)}) \\
&= \mathbf{x}^{(0)} - \alpha \nabla f(\mathbf{x}^{(0)}) - \alpha \nabla f(\mathbf{x}^{(1)}) \\
&\vdots
\end{aligned}$$

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(0)} - \alpha \sum_{i=1}^{t} \nabla f(\mathbf{x}^{(i)})$$

First Order Acceleration Methods

Let us try to unroll gradient descent:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(0)} - \alpha \sum_{i=1}^{t} \nabla f(\mathbf{x}^{(i)})$$

With momentum:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(0)} + \alpha \sum_{i=1}^{t} \frac{1 - \lambda^{t+1-i}}{1 - \lambda} \nabla f(\mathbf{x}^{(i)})$$

The more general form:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(0)} + \alpha \sum_{i=1}^{t} \gamma_i^t \nabla f(\mathbf{x}^{(i)}) \quad \text{for some } \gamma_i$$

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The more general form:

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(0)} + \alpha \sum_{i=1}^{t} \Gamma_i^t \nabla f(\mathbf{x}^{(i)}) \quad \text{for some diag. matrix } \Gamma_i$$

generalizes optimization algorithms like ADAM, AdaGrad, etc.

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Gradient Descent for Deep Learning

Gradient descent can be applied to nonconvex problems, without optimality guarantees.

In order to gain generalization, the following consideration is crucial:

We are rarely interested in the global optimum.

Gradient Descent for Deep Learning

Gradient descent can be applied to nonconvex problems, without optimality guarantees.

In order to gain generalization, the following consideration is crucial:

We are rarely interested in the global optimum.

Even for convex problems like:

- Linear regression ($\mathbf{X}$ can be huge and must be inverted/factorized)
- Logistic regression (no closed form solution)

We get more efficient and numerically stable solutions.

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Gradient Descent for Deep Learning

In the general DL setting:

Each parameter gets updated so as to decrease the loss:

$$\boldsymbol{\theta}_i \leftarrow \boldsymbol{\theta}_i - \alpha \frac{\partial \ell}{\partial \boldsymbol{\theta}_i}$$

The gradient tells us how to modify the parameters.

- $\boldsymbol{\theta}$ stores the neural network parameters, possibly millions
- The loss can be non-convex and non-differentiable
  Cannot even apply gradient descent!
- Must address computational complexity and numerical stability

Very often in DL we confront problems that have well defined solutions in mathematical terms, but require extra care and tweaking in practice.

Stochastic Gradient Descent

Recall that the loss is usually defined over n training examples:

$$\ell_\Theta(\{x_i, y_i\}) = \frac{1}{n}\sum_{i=1}^{n}(y_i - f_\Theta(x_i))^2$$

$$\ell_\Theta(\{x_i, y_i\}) = \frac{1}{n}\sum_{i=1}^{n}\hat{\ell}_\Theta(\{x_i, y_i\})$$

which requires computing the gradient for each term in the summation:

$$\nabla\ell_\Theta(\{x_i, y_i\}) = \frac{1}{n}\sum_{i=1}^{n}\nabla\hat{\ell}_\Theta(\{x_i, y_i\})$$

Two bottlenecks make gradient descent impractical:

- Number of examples
- Number of parameters

Stochastic Gradient Descent

$$\ell_\Theta(\{x_i, y_i\}) = \frac{1}{n}\sum_{i=1}^{n} \hat{\ell}_\Theta(\{x_i, y_i\})$$

$$\nabla \ell_\Theta(\mathcal{T}) = \frac{1}{n}\sum_{i=1}^{n} \nabla \hat{\ell}_\Theta(\mathcal{T})$$

Compute $\nabla \ell_\Theta$ for a small representative subset of $m \ll n$ examples:

$$\frac{1}{m}\sum_{i=1}^{m} \nabla \hat{\ell}_\Theta(\mathcal{B}) \approx \frac{1}{n}\sum_{i=1}^{n} \nabla \hat{\ell}_\Theta(\mathcal{T})$$

The mini-batch $\mathcal{B} \subset \mathcal{T}$ is drawn uniformly.

The true gradient $\nabla \ell_\Theta$ is approximated, but with a significant speed-up.

Example: MNIST dataset

$$n = 60{,}000\,,\;\; m = 10 \quad \Rightarrow \quad 6{,}000 \times \;\text{speedup}$$

Stochastic Gradient Descent

The algorithm is as follows:

1. Initialize $\boldsymbol{\theta}$.
2. Pick a mini-batch $\mathcal{B}$.
3. Update with the downhill step (use momentum if desired):

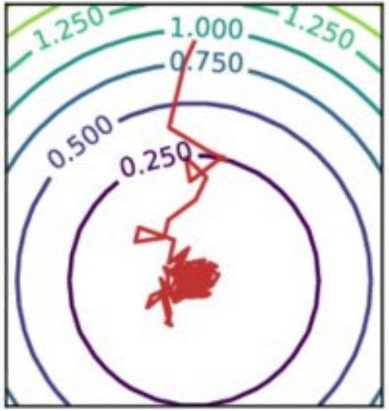
$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \nabla \ell_{\boldsymbol{\theta}}(\mathcal{B})$$

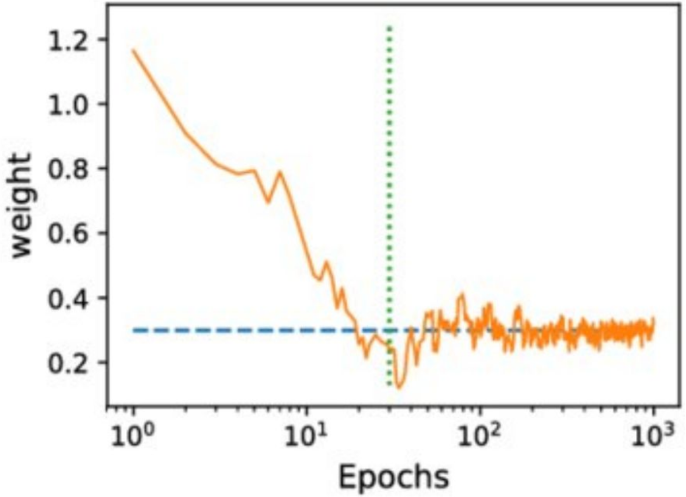
4. Go back to step (2).

When steps (2)-(4) cover the entire training set $\mathcal{T}$ we have an epoch.
Like gradient descent, the algorithm proceeds for many epochs.

Remark: The update cost is constant regardless of the size of $\mathcal{T}$.

Stochastic Gradient Descent





SGD does not stop at the minimum.

Oscillations are due to the noise induced by the random sampling.

Asymptotic upper bounds

For a convex problem, consider the inequality:

$$|\underbrace{\ell(\mathsf{h}_\Theta)}_{\text{GD/SGD}} - \underbrace{\ell(f^*)}_{\text{true}}| < \underbrace{\rho}_{\text{accuracy}}$$

n training examples
d parameters
κ, ν are constants related to the conditioning of the problem

	cost per iteration	**iterations to reach ρ**
GD	$O(nd)$	$O(\kappa \log \frac{1}{\rho})$
SGD	$O(d)$	$\frac{\nu\kappa^2}{\rho} + o(\frac{1}{\rho})$

SGD does not depend on the number of examples, implying better generalization

Bottou and Bousquet, "The Tradeoffs of Large Scale Learning", NIPS 2008

Stochastic Gradient Descent

Some practical considerations:

- Needs the training data to be shuffled to avoid data bias.

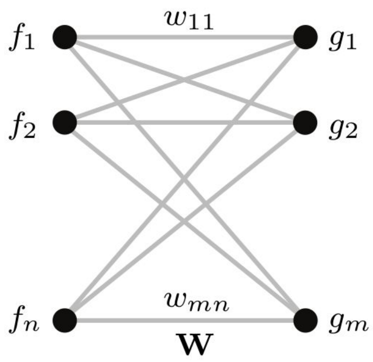
$$\frac{1}{m}\sum_{i=1}^{m}\nabla\hat{\ell}_{\Theta}(\mathcal{B}) \not\approx \frac{1}{n}\sum_{i=1}^{n}\nabla\hat{\ell}_{\Theta}(\mathcal{T})$$

- Each mini-batch can be processed in parallel.
  Batch size is limited by hardware.
- Large datasets: more steps to reach convergence.
  Remarkably rapid initial progress and slow asymptotic convergence.
- Small mini-batches can offer a regularizing effect.
  Very small batches $\rightarrow$ high variance in the estimation of the gradient
  $\rightarrow$ use a small learning rate to maintain stability.

SGD can find a low value of the loss quickly enough to be useful, even if it's not a minimum.

Slide Credit: E. Rodolà

Neural Network



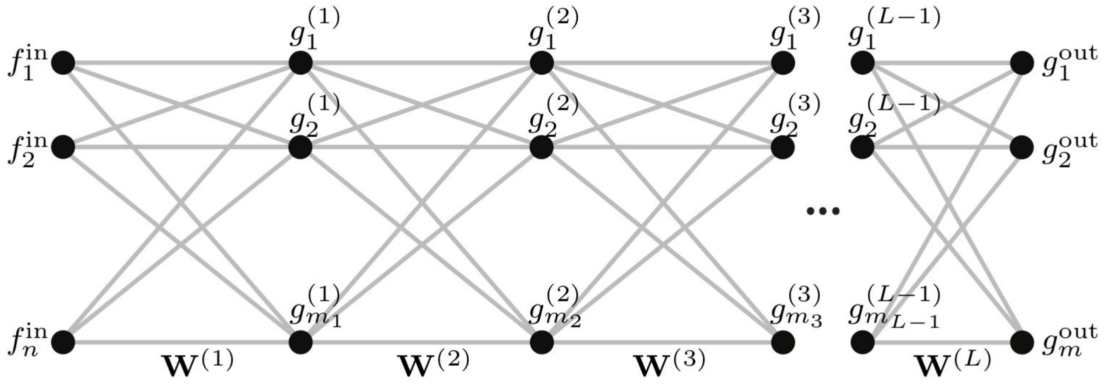
Single linear layer

Linear layer $\quad g_\ell = \sigma\left(\sum_{\ell'=1}^{n} f_{\ell'} w_{\ell,\ell'}\right) \quad \begin{array}{l}\ell = 1,\ldots,m \\ \ell' = 1,\ldots,n\end{array}$

Activation, e.g. $\quad \sigma(x) = \max\{x, 0\}$ $\quad$ rectified linear unit (ReLU)

Parameters $\quad$ layer weights $\mathbf{W}$ (including bias)

Neural Network



Deep neural network consisting of L layers

Linear layer $$g_\ell^{(k)} = \sigma\left(\sum_{\ell'=1}^{m_{k-1}} g_{\ell'}^{(k-1)} w_{\ell,\ell'}^{(k)}\right) \qquad \begin{array}{l}\ell = 1, \dots, m_k \\ \ell' = 1, \dots, m_{k-1}\end{array}$$

Activation, e.g. $\sigma(x) = \max\{x, 0\}$ rectified linear unit (ReLU)

Parameters weights of all layers $\mathbf{W}^{(1)}, \dots, \mathbf{W}^{(L)}$ (including biases)

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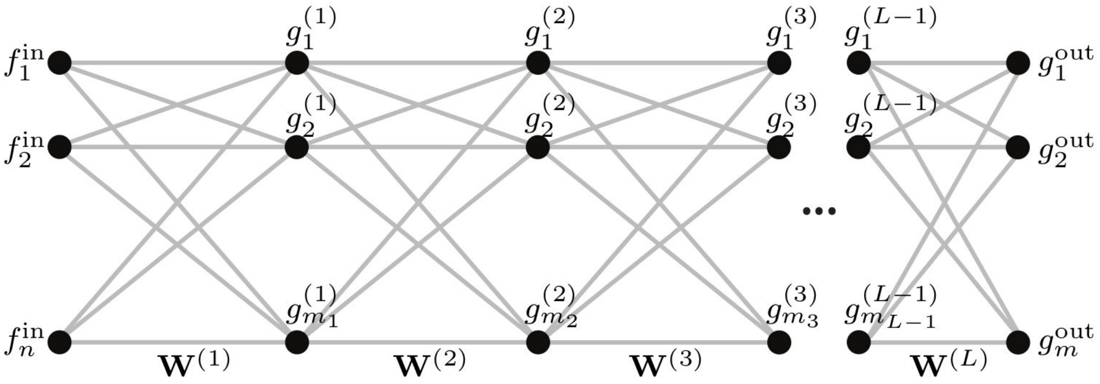
Neural Network



Deep neural network consisting of L layers

Linear layer $\quad \mathbf{g}^{(k)} = \sigma\left(\mathbf{W}^{(k)}\mathbf{g}^{(k-1)}\right)$

Activation, e.g. $\quad \sigma(x) = \max\{x, 0\}$ rectified linear unit (ReLU)

Parameters $\quad$ weights of all layers $\mathbf{W}^{(1)}, \ldots, \mathbf{W}^{(L)}$ (including biases)

The need of priors

Deep feed-forward networks are provably universal.

However:

- We can make them arbitrarily complex.
- The number of parameters can be huge.
- Very difficult to optimize.
- Very difficult to achieve generalization.